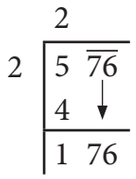
Step 3:

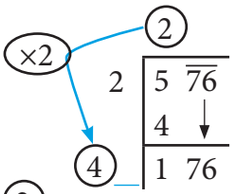
Bring 76 down and write it down to the right of the remainder 1.

Now, the new dividend is 176.



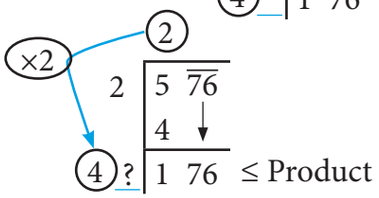
Step 4:

To find the new divisor, multiply the earlier quotient (2) by 2 (always) and write it leaving a blank space next to it.



Step 5:

The new divisor is 4 followed by a digit. We should choose this digit next to 4 such that the new quotient multiplied by the new divisor will be less than or equal to 176.



Step 6:

Clearly, the required digit here has to be 4 or 6. (Why?)

When we calculate, $46 \times 6 = 276$ whereas $44 \times 4 = 176$.

Therefore, we put 4 in the blank space and write $44 \times 4 = 176$ below 176 and subtract to get the remainder 0 and the quotient at the top, that is 24 is the square root of 576.

$\therefore \sqrt{576} = 24$.

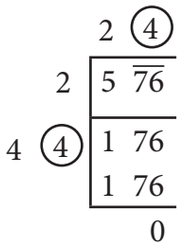
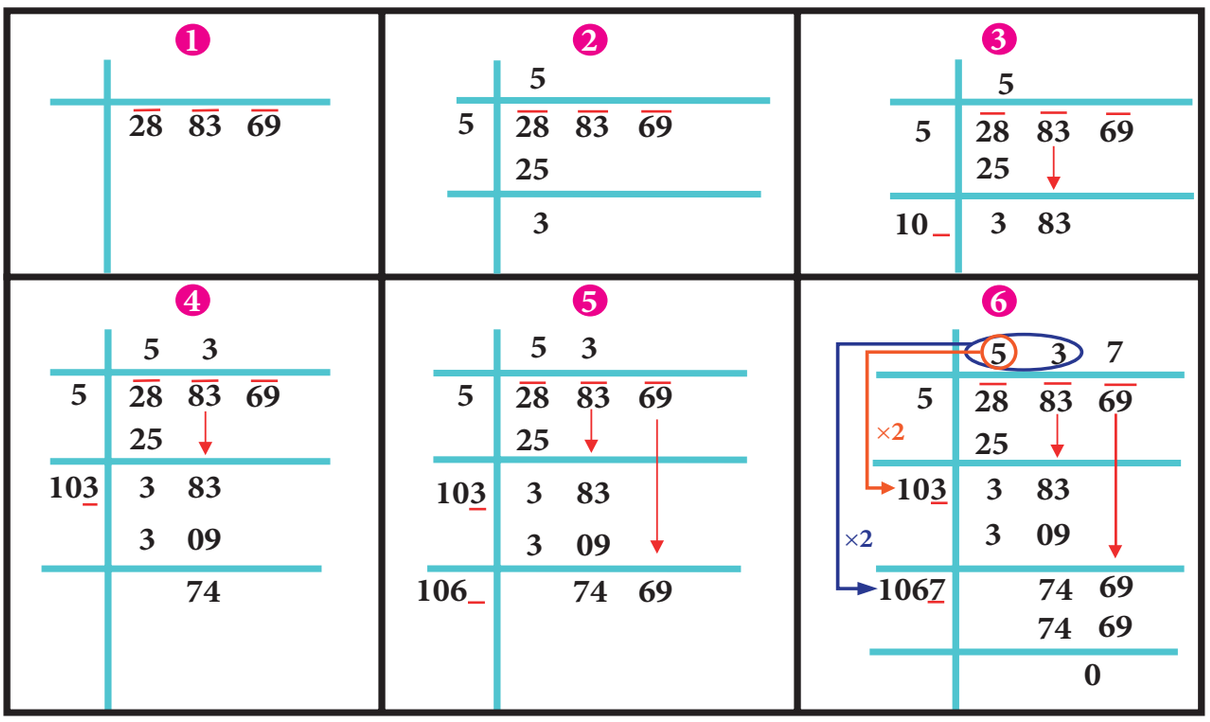


Illustration 2

In the following example, follow the figures one after another and try to understand what each figure explains, the stage by stage and the gradual computation of computing the square root of 288369.



We find that $\sqrt{288369} = 537$.

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